DESCENT DATA FOR QUASI-COHERENT SHEAVES OVER A STANDARD FPQC COVERING OF AN AFFINE SCHEME ARE EFFECTIVE

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file descent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is introduced by the author as a restatement of the cited proposition in terms of schemes, and then shows the restatement in detail; the word 'restatement' is the author's and the argument is displayed rather than assumed.

Editorial note. The counted spans carry 2 navigation stubs; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- Standard fpqc coverings of an affine scheme. Topologies, Definition topologies-definition-standard-fpqc, invoked at proof lines 1770-1771. Fixes the hypotheses, in particular that the family is finite, surjective and flat.
- Effectivity of descent for modules along a faithfully flat ring map. Same chapter, Proposition proposition-descent-module, invoked at proof lines 1780 and 1796-1797. Supplies the module statement to which the scheme statement is reduced.
- Descent data for a family of morphisms versus for the induced morphism from the disjoint union. Same chapter, the discussion identifying descent data for a family with descent data for the coproduct morphism, used at proof lines 1781-1786. Reduces the finite family to a single morphism and carries effectivity along.

Source reference. Original proposition 023N: sources/stacks-descent/pinned-original-source.tex, source lines 859–859. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original definition 022F: sources/stacks-chow/tags.txt, source lines 2704–2704. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Date: stacks-project:descent.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let S be an affine scheme. Let $\mathcal{U} = \{f_i : U_i \to S\}_{i=1,\ldots,n}$ be a standard fpqc covering of S, see Topologies, Definition 022F. Any descent datum on quasi-coherent sheaves for $\mathcal{U} = \{U_i \to S\}$ is effective. Moreover, the functor from the category of quasi-coherent $\mathcal{O}_S$-modules to the category of descent data with respect to $\mathcal{U}$ is fully faithful.*

Proof. This is a restatement of Proposition 023N in terms of schemes. First, note that a descent datum ξ for quasi-coherent sheaves with respect to $\mathcal{U}$ is exactly the same as a descent datum ξ' for quasi-coherent sheaves with respect to the covering $\mathcal{U}' = \{\coprod_{i=1,\ldots,n} U_i \to S\}$. Moreover, effectivity for ξ is the same as effectivity for ξ'. Hence we may assume $n = 1$, i.e., $\mathcal{U} = \{U \to S\}$ where U and S are affine. In this case descent data correspond to descent data on modules with respect to the ring map

$$\Gamma(S, \mathcal{O}) \longrightarrow \Gamma(U, \mathcal{O}).$$

Since $U \to S$ is surjective and flat, we see that this ring map is faithfully flat. In other words, Proposition 023N applies and we win. □